

Skills Summary

Binomial Coefficients in Expansion and Probability

Use this reference while completing your worksheet or when studying.

1. Evaluating a binomial coefficient

$\binom{n}{k} = \frac{n!}{k!(n-k)!}$ — the number of ways to choose k things from n when order does not matter. Cancel first, multiply last: $\binom{n}{k}$ is k factors falling from n , over $k!$. Two facts worth memorising: $\binom{n}{k} = \binom{n}{n-k}$ (choosing what to take is choosing what to leave), and $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$, which is Pascal's rule and the reason the triangle works.

Example: Evaluate $\binom{6}{2}$.

Step 1. The order of the two chosen items does not matter, so this is a combination — use the factorial formula and cancel before multiplying.

Step 2. $\binom{6}{2} = \frac{6!}{2!4!} = \frac{6 \cdot 5}{2 \cdot 1} \cdot \binom{6}{2} = 15$

Try it: Evaluate $\binom{8}{3}$.

check: 56

2. Expanding with the binomial theorem

$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$. Read a term off in two pieces: the binomial coefficient, then the constants raised to their powers. The coefficient of a term is the *product* of the two — $\binom{n}{k}$ alone is the answer only when both a and b are x and 1. Watch the signs: a negative b makes every odd- k term negative.

Example: Expand $(x + 3)^3$.

Step 1. A small whole power of a binomial, so read the coefficients straight off row 3 of Pascal's triangle rather than multiplying three brackets.

Step 2. Row 3 is 1, 3, 3, 1, so the terms are x^3 , $3x^2(3) = 9x^2$, $3x(3)^2 = 27x$, and $(3)^3 = 27$.
 $(x + 3)^3 = x^3 + 9x^2 + 27x + 27$

Try it: Expand $(x - 2)^3$.

check: $x^3 - 6x^2 + 12x - 8$

3. Counting your way to a probability

When every outcome is equally likely, $P = \frac{\text{favourable}}{\text{total}}$ and both counts are usually binomial coefficients. Drawing without replacement: count the favourable draws by choosing from each group separately and *multiplying*, then divide by one coefficient for all the draws. Always count both numbers the same way — if the total ignores order, the favourable count must too.

Example: A box has 3 defective and 7 good parts. Four are drawn at once. Find the probability that exactly 1 is defective.

Step 1. The draws are all at once and order does not matter, so count outcomes rather than using a per-trial probability formula.

Step 2. Favourable: $\binom{3}{1}\binom{7}{3} = 3 \cdot 35 = 105$. Total: $\binom{10}{4} = 210$. $P = \frac{105}{210} = \frac{1}{2}$

Try it: A bag has 3 red and 5 green counters; 2 are drawn at once. Find the probability both are red.

$\frac{87}{8}$:xæp

4. Binomial probability: repeated independent trials

For n independent trials each succeeding with probability p , $P(\text{exactly } k) = \binom{n}{k}p^k(1-p)^{n-k}$. The powers price *one* sequence of k successes and $n - k$ failures; the coefficient counts how many such sequences there are. Use this only when the trials are independent with a fixed p — drawing without replacement changes p as you go, and needs section 3 instead.

Example: A fair die is rolled 4 times. Find the probability of exactly 2 sixes.

Step 1. The rolls are independent and the chance of a six is fixed at $\frac{1}{6}$, so this is the binomial formula, not a counting fraction.

Step 2. $\binom{4}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^2 = 6 \cdot \frac{1}{36} \cdot \frac{25}{36}$. $P = \frac{25}{216}$

Try it: A fair die is rolled 3 times. Find the probability of exactly 1 six.

$\frac{72}{27}$:xæp